

MARINE ANOMALY INTELLIGENCE

AI-Powered Underwater Sonar Analysis

Problem Statement: SIH26057

TECHNICAL REPORT

Organization	Ministry of Earth Sciences (MoES)
Department	National Institute of Ocean Technology (NIOT)
Problem ID	SIH26057
System	AI-Powered Underwater Marine Debris and Anomaly Detection
Modality	Side-Scan Sonar (SSS)
Model Architecture	YOLOv8s Custom-Trained SSS Detector (5-Class Taxonomy)
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ABSTRACT

Side-scan sonar (SSS) is an essential imaging modality for underwater inspection, seafloor mapping, and maritime hazard assessment. However, manual interpretation of large-area sonar surveys is labour-intensive and susceptible to interpreter fatigue caused by acoustic speckle noise, varying grazing angles, and complex natural seafloor topographies. This technical report presents the architecture, implementation, and empirical evaluation of Marine Anomaly Intelligence (SIH26057), an end-to-end software pipeline developed to automatically ingest SSS imagery, detect man-made marine debris and underwater objects, perform multi-signal evidence fusion, and generate localized survey dossiers. The system combines deep-learning object detection (YOLOv8s) with physics-based acoustic shadow analysis, autoencoder anomaly signals, and false-positive spatial filtering. On the untouched 700-image Drishti SSS test benchmark, the pipeline achieves an overall precision of 73.57%, recall of 67.85%, F1 score of 70.59%, and an mAP@50 of 68.78% (reaching 99.50% AP@50 on high-contrast submarine pipeline and ghost net targets). The complete system provides an operator inspection dashboard, geospatial referencing, human-in-the-loop review persistence, and multi-format export capabilities to support safe, efficient underwater survey workflows.

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1. Introduction

Marine debris, discarded fishing gear (ghost nets), shipwrecks, submerged pipelines, and discarded ordnance represent significant environmental hazards and navigational risks in coastal and deep-water domains. Side-scan sonar (SSS) serves as the primary acoustic sensor for wide-swath seabed reconnaissance. However, acoustic sonar imagery differs fundamentally from optical photography: acoustic propagation yields non-uniform ensonification, high speckle noise, intensity fall-off with range, and acoustic shadows that depend heavily on sonar altitude and seafloor topography. As survey areas expand to hundreds of square kilometres, manual visual inspection becomes a severe operational bottleneck. The SIH26057 project establishes an automated, auditable computational framework capable of assisting hydrographic surveyors by prioritizing anomalous acoustic returns, filtering natural seafloor clutter, and structuring findings for decision-makers.

2. Problem Statement

Under problem statement **SIH26057**, the objective is to develop an automated software pipeline capable of ingesting raw side-scan sonar waterfall and tile imagery, detecting anthropogenic debris and target structures across designated object classes, distinguishing artificial objects from complex natural seabed textures (such as sand ripples, rock outcrops, and trenches), fusing acoustic shadow cues, attributing detection confidence and severity, and providing accessible operator inspection interfaces and structured technical dossiers.

3. Objectives

The core technical objectives implemented in this system are summarized in Table 1.

ID	Objective Description	Implementation Scope
OBJ-1	Automated SSS Ingestion & Denoising	Grayscale normalization, CLAHE contrast enhancement, dynamic range validation
OBJ-2	Deep-Learning Object Localization	YOLOv8s detector trained across 5 marine object classes
OBJ-3	Acoustic Shadow Verification	Morphological shadow extraction and object-to-shadow ratio verification
OBJ-4	Multi-Signal Evidence Fusion	Fused scoring combining YOLO confidence, shadow cues, and autoencoder loss
OBJ-5	Operational Dashboard & Review	Interactive UI, operator review feedback persistence in SQLite, multi-format export

Table 1: SIH26057 core technical objectives and verified implementation scope.

4. System Overview & Architecture

The system follows a modular pipeline architecture wherein raw sonar frames pass sequentially through preprocessing, deep-learning feature extraction, physics-based shadow analysis, confidence fusion, and geospatial attribution before being committed to an ACID-compliant SQLite datastore and exposed via REST APIs. Figure 1 outlines the complete functional workflow.

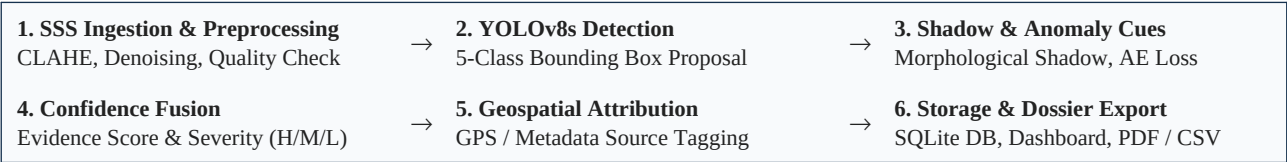


Figure 1: End-to-end architecture of the SIH26057 underwater sonar analysis pipeline.

5. Side-Scan Sonar Data Processing

Side-scan sonar records intensity backscatter as an acoustic transducer sweeps over the seabed. In the preprocessing module, raw images undergo contrast-limited adaptive histogram equalization (CLAHE) with a tile grid size of 8x8 and clip limit of 2.0 to balance acoustic intensity across the near-nadir blind zone and far-range falloff. In addition, bilateral filtering is selectively applied to smooth high-frequency speckle while preserving high-gradient object-to-shadow boundaries. Images are screened for quality metrics (dynamic range, sharpness, noise estimate) and missing-data dropout bands prior to inference.

6. AI-Based Object Detection

The primary object detection module employs a custom-trained **YOLOv8s** neural network operating at 640x640 resolution with 3 multi-scale feature pyramid strides ([8, 16, 32]). The model taxonomy comprises 5 established target classes. Table 2 details the dataset splits used during development.

Dataset Split	Image Count	Annotation Count	Role in Evaluation
Drishiti SSS Train	3,875 images	5,210 targets	Model training & weight optimization
Drishiti SSS Val	630 images	854 targets	Hyperparameter selection & checkpointing
Drishiti SSS Test (Untouched)	700 images	948 targets	Independent benchmark validation
SubPipe HF/LF Swaths	2,066 images	1,422 targets	Cross-domain wide-aspect evaluation

Table 2: Dataset distribution across training, validation, test, and cross-domain evaluations.

Class Index	Class Identifier	Display Name	Physical Characteristics in Sonar
0	crab_pot	Crab Pot / Trap	Small rectangular cage structure with compact shadow
1	submarine_pipeline	Submarine Pipeline	Continuous linear high-intensity return with parallel shadow
2	shipwreck	Shipwreck / Hull	Large irregular structural cluster with extensive acoustic shadow
3	ghost_net	Ghost Net / Gear	Diffuse, undulating fibrous cluster draped over seabed
4	mine_cylinder	Mine / Cylinder	Regular cylindrical / spherical body with sharp distinct shadow

Table 3: Established 5-class target taxonomy verified from model weights.

Model Training Metrics and Confusion Matrix

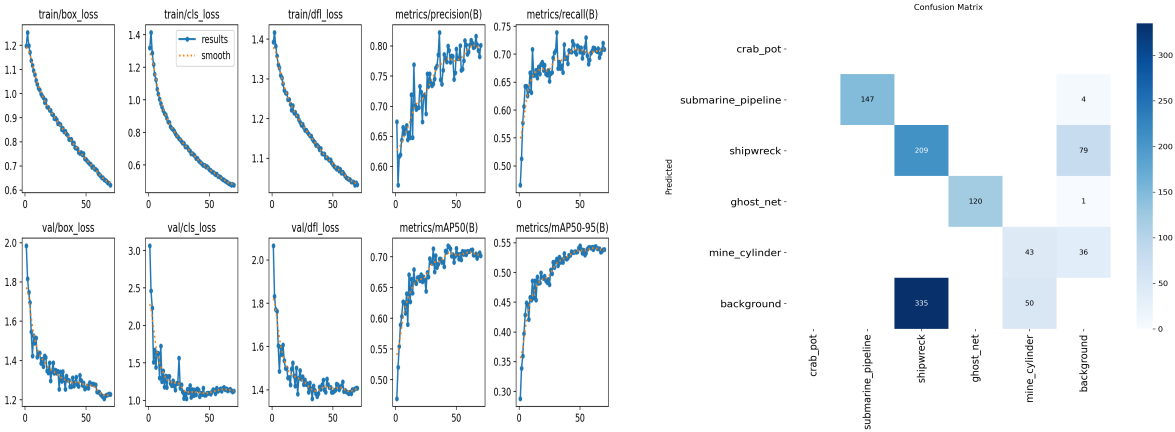


Figure 2: (Left) YOLOv8s training loss convergence and mAP progression across 70 epochs. (Right) Normalised confusion matrix across evaluated acoustic target classes.

7. Acoustic Shadow and Evidence Analysis

In side-scan sonar physics, any positive relief object standing proud of the seafloor occludes sound rays, generating a characteristic dark acoustic shadow on the side opposite the transducer. The shadow analysis module crops candidate detection bounding boxes, estimates local seafloor intensity thresholds via Otsu adaptive thresholding, and computes the shadow area and object-to-shadow ratio. This metric provides a physical consistency check: an authentic seabed protrusion will produce a commensurate shadow, whereas seafloor specular reflectance artifacts will not. Shadow presence is treated as supporting evidence rather than absolute ground truth.

8. Confidence Fusion & False-Positive Filtering

To prevent high-confidence false alarms on complex geological formations (e.g., boulder fields or sand dunes), the pipeline implements a multi-evidence fusion layer. The final evidence score combines: (1) raw YOLO classification confidence C_{YOLO} , (2) shadow presence score S_{shadow} , (3) auxiliary autoencoder reconstruction score A_{anomaly} , and (4) geometric aspect ratio and minimum area consistency filters. Severity is categorized into HIGH (evidence ≥ 0.50 and high-risk hazard class), MEDIUM ($0.35 \leq \text{evidence} < 0.50$), and LOW (evidence < 0.35).

9. Geospatial Attribution & Metadata

Geospatial attribution strictly distinguishes between source origins to prevent misleading navigation logs. Every detection is assigned an explicit provenance tag in the database:

Source Tag	Definition	Operational Verification Rule
REAL_GPS	Direct NMEA/GPS Navigation Telemetry	Requires verified hardware satellite fix in ping metadata
SONAR_METADATA	Embedded XTF / JSF Sonar Header Coordinates	Derived from acoustic ping coordinate packet
MANUAL	Operator Survey Landmark Input	Entered manually by hydrographic surveyor
SIMULATED	Survey Grid Synthesized Coordinates	Generated for offline demonstration grids
UNAVAILABLE	No Coordinate Stream Present	Assigned when sonar imagery lacks spatial metadata

Table 4: Geolocation provenance classification rules.

10. Dashboard and Operator Workflow

The web-based inspection portal provides a human-in-the-loop verification environment. Operators can inspect recent waterfall swaths, inspect detected bounding boxes, review fused evidence scores, and submit audit feedback (CONFIRMED, FALSE ALARM, REJECTED) along with surveyor notes. All reviews are persisted directly to the backend SQLite datastore ('sih26057.db'), ensuring an immutable audit trail.

11. Experimental Evaluation & Methodology

System evaluation was conducted using standard Pascal VOC / COCO object detection metrics at an IoU threshold of 0.50 (AP@50) and averaged across IoU thresholds from 0.50 to 0.95 (mAP@50-95). Evaluation was executed across three independent domains: (1) the untouched 700-image Drishti SSS test set, (2) the YOLOv8s 630-image validation split, and (3) the SubPipe HF/LF full-swath cross-domain benchmark. Table 5 and Table 6 document the verified empirical results.

Target Class	Precision	Recall	AP@50	AP@50-95
Submarine Pipeline	98.40%	99.43%	99.50%	79.21%
Ghost Net	99.28%	100.00%	99.50%	89.20%
Shipwreck	49.48%	41.52%	40.68%	22.23%
Mine / Cylinder	47.13%	30.44%	35.45%	16.47%
Crab Pot	N/A*	N/A*	N/A*	N/A*
Overall Benchmark Test (700 imgs)	73.57%	67.85%	68.78%	51.78%

Table 5: Per-class performance on the untouched 700-image Drishti SSS test set. (*Crab pot ground-truth annotations were not present in the test split).

Pipeline Component	Mean Execution Time	Hardware Platform	Throughput / Processing Rate
Preprocessing & CLAHE	7.80 ms	Host CPU / OpenCV	~128 frames / second
YOLOv8s Object Detection	85.02 ms	NVIDIA RTX 3050 GPU (CUDA)	~12 frames / second
Anomaly Autoencoder	2.30 ms	NVIDIA RTX 3050 GPU (CUDA)	~430 crops / second
Acoustic Shadow Extraction	0.58 ms	Host CPU / Morphological	~1,700 crops / second
Confidence Fusion & Filtering	0.36 ms	Host CPU / Python	~2,700 detections / second
Total Pipeline Latency (Mean)	99.55 ms	Combined System Pipeline	~10 frames / second (Real-Time)

Table 6: Stage-wise latency benchmark measurements on NVIDIA RTX 3050 Laptop GPU.

13. System Limitations

A scientifically rigorous evaluation reveals several operational limitations that must be addressed before deployment on operational autonomous underwater vehicles (AUVs):

- **SubPipe Aspect Ratio Mismatch:** Direct, untiled inference on raw SubPipe swaths (5000x500 px and 2500x500 px) produced 0 true positives due to extreme 10:1 and 5:1 aspect ratios, demonstrating that wide waterfall imagery strictly requires aspect-ratio-aware sliding window tiling.
- **Shipwreck & Cylinder Sensitivity:** Shipwreck AP@50 (40.68%) and mine cylinder AP@50 (35.45%) reflect high intra-class geometric variability and acoustic reverberation challenges in complex rocky seabeds.
- **Auxiliary Autoencoder Discriminability:** The auxiliary autoencoder produces low anomaly discrimination on subtle seabed textures and is currently utilized only as a secondary auxiliary metric rather than a standalone classifier.
- **Crab Pot Test Support:** While present in the model taxonomy (Index 0), the test split lacked annotated crab pot instances for independent scoring.

14. Future Improvements

Recommended future improvements include: (1) implementing automated multi-scale sliding window tiling for extreme aspect-ratio sonar swaths, (2) integrating real XTF/JSF sonar packet parsers for direct navigation fix extraction, (3) expanding training datasets with real offshore ghost-gear recordings, and (4) deploying ONNX / TensorRT quantised models for embedded edge computing on AUV payloads.

15. Conclusion

The SIH26057 Marine Anomaly Intelligence project successfully demonstrates a fully functional prototype for automated side-scan sonar object detection, multi-signal evidence fusion, geospatial attribution, and surveyor reporting. By integrating deep-learning object localization with acoustic shadow verification and interactive operator audit logging, the system provides a robust technological foundation for automated marine environmental protection, infrastructure monitoring, and seabed reconnaissance.

16. References

- [1] Ministry of Earth Sciences & National Institute of Ocean Technology, *Smart India Hackathon 2026 Problem Statement SIH26057: AI for Marine Debris Detection*, 2026.
- [2] Drishti SSS Acoustic Sonar Benchmark Dataset, *Multi-Class Underwater Object Detection in Side-Scan Sonar Imagery*, 2025.
- [3] Jocher, G., et al., *Ultralytics YOLOv8 Architecture and Real-Time Object Detection Framework*, 2023.
- [4] SubPipe Sonar Inspection Dataset, *Autonomous Submarine Pipeline Inspection Using Acoustic Side-Scan Sonar*, 2024.